Relational Calculus — Music Library

txt2tex example

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$$[TrackId, ArtistId, AlbumId, PlaylistId, GenreName, TrackTitle, AlbumTitle, ArtistName]$$

Track

$tid : TrackId$
$title : TrackTitle$
$alid : AlbumId$
$duration : \mathbb{N}$
$genre : GenreName$

Artist

$arid : ArtistId$
$name : ArtistName$

Album

$alid : AlbumId$
$title : AlbumTitle$
$arid : ArtistId$

$member : PlaylistId \leftrightarrow TrackId$

$Jazz : GenreName$

Query 1: Single-relation selection

Retrieve the title and genre of every jazz track. This is a plain single-relation query over ‘Track‘ — no join required.

$$\{\, t : Track \mid t.genre = Jazz \bullet \\ \quad \langle\!| title == t.title, genre == t.genre |\!\rangle \,\}$$

Query 2: Two-relation join via comprehension

For each track, return its title together with the name of the artist who recorded the album it belongs to. Joins ‘Track‘ and ‘Album‘ on ‘AlbumId‘, then ‘Album‘ and ‘Artist‘ on ‘ArtistId‘.

$$\{\, t : Track; al : Album; ar : Artist \mid t.alid = al.alid \land \\ \quad al.arid = ar.arid \bullet \\ \quad \langle\!| track == t.title, artist == ar.name |\!\rangle \,\}$$

Query 3: Three-relation chain with playlist

List every track title that appears on at least one playlist, together with the album title. Chains ‘member‘, ‘Track‘, and ‘Album‘ through three declarations.

$$\{\, p : \mathit{PlaylistId};\ t : \mathit{Track};\ al : \mathit{Album} \mid (p, t.\mathit{tid}) \in \mathit{member} \land \\ t.\mathit{alid} = al.\mathit{alid} \bullet \\ \lblot\, \mathit{playlist} == p, \mathit{track} == t.\mathit{title}, \mathit{album} == al.\mathit{title} \,\rblot \,\}$$

Query 4: Binding output, filtered by duration

Return a binding of ‘(track title, album title)‘ for every track whose duration exceeds 300 seconds. The characteristic expression is a two-component binding; no plain variable appears in the output.

$$\{\, t : \mathit{Track};\ al : \mathit{Album} \mid t.\mathit{alid} = al.\mathit{alid} \land \\ t.\mathit{duration} > 300 \bullet \\ \lblot\, \mathit{track} == t.\mathit{title}, \mathit{album} == al.\mathit{title} \,\rblot \,\}$$

Query 5: Multi-decl forall

Assert that every track on any playlist belongs to a known album. Universal quantifier over two declarations with a conjunction predicate over projections.

$$\forall\, p : \mathit{PlaylistId};\ t : \mathit{Track} \mid \\ (p, t.\mathit{tid}) \in \mathit{member} \bullet t.\mathit{alid} \in \{\, al : \mathit{Album} \bullet al.\mathit{alid} \,\}$$

Query 6: Multi-decl comprehension, no characteristic expression

Collect the set of ‘(Track, Album)‘ schema pairs for long tracks. When the characteristic expression is omitted Z defaults to the signature tuple (Z RM §3.9). The Q2(d) parser fix makes the dot-separator optional.

$$\{\, t : \mathit{Track};\ al : \mathit{Album} \mid t.\mathit{alid} = al.\mathit{alid} \land \\ t.\mathit{duration} > 300 \,\}$$